



The ProbLemma Channel Recurrence Relations

This document links all the recurrence relations evaluated on the ProbLemma channel to the episodes in which these evaluations were carried out.



In addition, all the above episodes are collected in [the “Recurrence Relations Are Us” play list](#).

1. Season 3 Episode 41 (S3M22): How Generating Functions Can Solve Linear Non-homogeneous Recurrence Relations

$$t_n = 6t_{n-1} - t_{n-2} - 4, \quad t_0 = 2, t_1 = 4$$

An analytic representation of a generating ordinary power series function for the above recurrence relation is:

$$G(x) = \frac{1}{1-x} - \frac{3+2\sqrt{2}}{2(x-3-2\sqrt{2})} - \frac{3-2\sqrt{2}}{2(x-3+2\sqrt{2})}$$

and its power series decomposition is:

$$G(x) = \sum_{n=0}^{+\infty} \left[1 + \frac{1}{2} (3-2\sqrt{2})^n + \frac{1}{2} (3+2\sqrt{2})^n \right] x^n$$

As such:

$$t(n) = 1 + \frac{1}{2} (3-2\sqrt{2})^n + \frac{1}{2} (3+2\sqrt{2})^n$$

with the first few terms:

$$t_n = \{2, 4, 18, 100, 578, 3364, \dots\}$$

This relation is also solved purely geometrically in [the “Inversive Geometry” course](#) in following episode: [IGRP Episode 20 Prctice 2. Problem 21. Inversion Meets A Recurrence Relation](#).

2. Season X Episode Y (SxMn): Lorem ipsum

3. Season X Episode Y (SxMy): Lorem Ipsum